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ORIGINAL PAPER

Assessment of Energy Consumption in a Meat-Processing Plant—a Case Study

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Abstract Meat-processing plants have their specific characteristics due to quantity of raw materials processed, production technology, degree of mechanisation of production and space utilization. These characteristics affect variability in consumption of energy carriers. This paper presents a case study of energy consumption in a small meat processing plant. It is attempting to identify and evaluate key factors which may be useful in conducting an in-depth analysis of energy consumption in the meat industry, and so this analysis may be important in selecting the best production techniques available. The results of the analysis are of critical importance for the selection of production technology selection assuming high plant efficiency.

Keywords Meat processing · Energy consumption · Energy efficiency

Abbreviations

a	Regression coefficient
A_c	Monthly thermal energy consumption ($A_c=B_{rz}Q^r_w$, for Q^r_w=43.1 MJ/kg of fuel)
A_e	Active electric energy monthly consumption kWh/month
A_{t1}	Total monthly energy consumption (after conversion 1 kWh=12 MJ), in megajoules per month
A_{t2}	Total monthly energy consumption (after conversion 1kWh=3,6MJ), in megajoules per month
b	Regression coefficient
B_{ce1}	Monthly coal equivalent consumption based on A_{t1}, in kilograms CE per month
B_{ce2}	Monthly coal equivalent consumption based on A_{t2}, in kilograms CE per month
B_{rz}	Monthly real fuel consumption, in kilograms per month
EE	Energy efficiency megagrams of product per kilowatt hour, megagrams of product per gigajoules
n	Total number of employees in the plant person
P	Total capacity of installed electrical equipment, in kilowatts
R^2	Determination coefficient ($R^2=r^2\times100$ %) in percent
U_{en}	Power of installed electrical equipment per employee, in kilowatts per person
Q_{ce}	Calorific value of coal equivalent (29.3076 MJ/kg CE)
Q^r_w	Calorific value of real fuel, in megajoules per kilogram of fuel
V_1	Volume of the plant production space, in cubic meters
V_2	Total volume of the plant, in cubic meters
W_A	Aggregate indicator of energy consumption per unit of physical output (aggregate indicator of specific energy consumption SEC_A)
W_P	Production indicator of energy consumption per unit of physical output (production indicator of specific energy consumption SEC_P)
W_T	Technological indicator of energy consumption per unit of physical output (technological indicator of specific energy consumption SEC_T)
W_Z	Plant indicator of energy consumption per unit of physical output (plant indicator of specific energy consumption SEC_Z)
Z	Monthly production, in megagrams of product per month

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Introduction

Meat processing is one of the branches of the food industry (see Grochowicz and Walczyński 2004; Jekayinfa and

Bamgboye 2006; Wang 2008; Klemeš et al. 2008; Neryng et al. 1990; Wilhite 2008) for discussion of some aspects of energy consumption in the food industry). The energy consumption is one of the key production factors in the industry. It has a great impact on the meat processing plant efficiency and the natural environment. Consumption of energy carriers in meat processing plants depends on many factors such as, for example, size and structure of output, production technology used, thermo-physical properties of raw materials, mechanisation of production processes and utilization of processing capacity (see Cierach et al. 2000; Tkacz et al. 2000;Tkacz and Iwaniak 2001; Markowski et al. 2004; Marcotte et al. 2008; Banach and Żywica 2010; Houska et al. 2003; Li et al. 2010; Campañone et al.2010; Gogate 2011; Norton and Sun 2008). Demand for energy carriers is taken into account, among others, in the EU's Integrated Pollution Prevention and Control Directive which determines a set of rules and requirements for effective protection of the environment and application of the best available production techniques (WS Atkins Int 1998; European Commission 2006; Ramirez et al. 2006; IFC—World Bank Group 2007). Moreover, modern methods of energy management in small manufacturing plants require creation of databases which are useful in analysing energy efficiency of food processing and supply of more detailed input data (Swords et al. 2008).

To assess energy efficiency, unit energy consumption indicators as well as energy efficiency indicators are most commonly used. Efficiency is defined as a result of production (industrial) activity and it is expressed as a quotient of the result obtained to the effort expended. Improvement in energy efficiency may involve a reduction in demand for energy carriers during their transformation, transmission, and final use. It may be due to changes in technology which are capable of providing a constant or a higher level of production. Such improvements may result in energy conservation, reducing consumption of natural resources, reducing emissions and limiting the amount of waste generated at each stage of meat production and processing (Ramirez et al. 2006; Pagan et al. 2002; Pimentel et al. 2008).

The available sources in literature show that production energy consumption is expressed using different units, and the reasons for variation in energy carriers' consumption in meat-processing plants of various sizes were not fully explained (WS Atkins Int 1998; European Commission 2006; Ramirez et al. 2006; Podfilipski and Żelazny 1985; Singh 1986; Budny et al. 1987; Fritzson and Berntsson 2006).

Table 1 presents the figures for energy consumption in the meat processing plants referred to in the literature. Different ranges of unit energy consumption indicators listed in Fig. 1 were taken into account. Energy efficiency in meat processing is given by using indicators which are difficult to compare because they do not relate to specific conditions found in meat processing plants. Table 1 does not include data from poultry-processing plants which can be found in Dróżdż (2010); Jekayinfa (2007); Wojdalski et al. (2009).

The aim of this study was to determine the energy consumption of a small-sized pork meat processing plant. This study aims to provide detailed data which could be useful in creating models of enterprises of this kind as energy users as well as for determining the relationship between production volume and energy carriers' consumption which serves to evaluate energy efficiency of production. The aim of the study is also correlated with the study (Bunse et al. 2011), where the authors indicate a gap between the needs for research reported by manufacturing plants and scientific publications currently on offer.

Materials and Methods

The research data were collected in a meat processing plant in Mazowieckie Voivodship, Poland, employing 73 people in total, among whom 33 were directly involved in production.

The plant did not slaughter animals. Pre-treated frozen carcasses were purchased from slaughter plants.

Monthly production (Z) during the study period ranged from 303 to 699 Mg. The annual plant production profile: meat products 78.4 % and unprocessed meat 21.6 % of total production.

The plant's production area was 994.28 m^2; the total volume of the plant was 6,336.6 m^3 of which the production space accounted for 78.1 %. The combined power of all installed electrical equipment in the plant (P) was 180 kW. Breakdown of power by type of electrical equipment in the plant is shown in Table 2.

Table 2 shows that the largest category on the power balance sheet was meat cutters and grinders (22 %).

The values of the indicator expressing the degree of mechanisation of the production operations (Table 4) measured in several 1-month intervals during study ranged from 0.257 to 0.594 kW/Mg of processed products. The plant used conventional technologies to produce meat, meat products, and raw and melted animal fat.

Meeting the goals of the study required, among others, using a model of the meat processing plant as the end-user of energy carriers (Fig. 1) and applying the plant indicators of energy carriers' consumption, as applied in the study (Wojdalski et al. 2010). These indicators cover all energy receivers associated with transformation and use of energy carriers in the plant as a whole. The indicators WA, WT and WP are less informative as far as energy consumption of the plant is concerned. A study of these indicators is not within the scope this paper. They are included however in Fig. 1 because some of them are cited in the literature and reported in Table 1.

Table 1 Available research data on energy carriers' consumption in the meat processing industry

Energy carriers	Symbols and units	Indicators		Average	Source
		Range	Figures		
Electric energy	W_e [kWh/Mg of meat batter]	W_Z	80–2,420	720	WS Atkins Int (1998)
	W_e [kWh/Mg of product]	W_Z	70–8,140	1,260	WS Atkins Int (1998)
		W_T	65.3–96.3	–	Neryng et al. (1990)
		W_A	44–112	–	Tkacz (2002)
		W_Z	269–279	–	Kowalczyk & Netter (2008)
	W_e [kWh/Mg of meat]	W_Z	37.0–52.0[a]	44.5[a]	Wojdalski et al. (2010)
	W_e [kJ/kg of meat batter]	W_A	4–20	–	Dolata (1992)
Thermal energy	W_c [GJ/Mg of meat batter]	W_Z	5.89–58.1	12.9	WS Atkins Int (1998)
	W_c [MJ/Mg of meat]	W_Z	970–1,110[a]	1,040[a]	Wojdalski et al. (2010)
	W_c [Mg of coal/Mg of meat batter]	W_Z	–	0.96	WS Atkins Int (1998)
	W_c [GJ/Mg of product]	W_Z	0–220	24.5	WS Atkins Int (1998)
		W_Z	3.14–4.27[e]	3.61[e]	Skonecki et al. (2001)
		W_Z	2.10–2.26	–	Kowalczyk & Netter (2008)
	W_c [kg of steam/Mg of product]	W_Z	800–900	–	European Commission (2006)
	W_c [MJ/unit]	W_T	31.8–63.0[a]	44.6[a]	Neryng et al. (1990)
		W_T	22.8–45.2[b]	32.6[b]	
Total energy	W_c [MJ/Mg dress carcass weight] for whole and chilled	W_Z	–	1,390[c]	Ramirez et al.(2006)
		W_Z		2,097[d]	
	W_{t2} [MJ/Mg dress carcass weight] for whole and frozen	W_Z	–	2,110[c]	
		W_Z		3,128[d]	
	W_{t2} [MJ/Mg dress carcass weight] for cut-up, deboned and chilled	W_Z	–	2,146[c]	
		W_Z		2,849[d]	
	W_{t2} [MJ/Mg dress carcass weight] for cut-up, deboned and frozen	W_Z	–	2,866[c]	
		W_Z		3,884[d]	
	W_{t1} [MJ/Mg of meat]	W_Z	1,420–1,720[a]	1,570[a]	Wojdalski et al. (2010)
	W_{t2} [MJ/Mg of meat]	W_Z	1,110–1,280[a]	1,195[a]	
	W_{t2} [MJ/Mg of HSCW]	W_Z	1,200–4,800[f]	–	Hansen et al. (2000)
Coal equivalent	W_{ce1} [kg CE/Mg of meat]	W_Z	48.4–58.7[a]	53.6[a]	Wojdalski et al. (2010)
	W_{ce2} [kg CE/Mg of meat]	W_Z	37.9–43.7[a]	40.8[a]	

[a] Pig slaughter

[b] Cattle slaughter

[c] Beef, veal and sheep

[d] Pork

[e] Energy supplied by water steam

[f] *HSCW* hot standard carcass weight

It should be noted that these research methods could be modified to include the issues concerning combined heat and power as presented, among others, in the papers by Bianchi et al. (2006) and Bieranowski and Klonowski (2005).

To assess energy efficiency of production in the analysed plant, we used the indicator method, which allows online analysis of the plant performance. It is possible, by comparison with results obtained in similar plants, to indicate directions of possible changes aimed at increasing energy efficiency. In order to characterise the energy management of the plant, the indicators contained in Table 3 were adopted. These indicators cover the W_Z range shown in Fig. 1. The indicators listed in rows 8, 10 and 13 take into account the efficiency (η=0.3) of electric energy generation and its transmission too.

From the cost and specific technology evaluation standpoints, it is important to use a value indicator which includes total energy consumption in the processing plant. To achieve

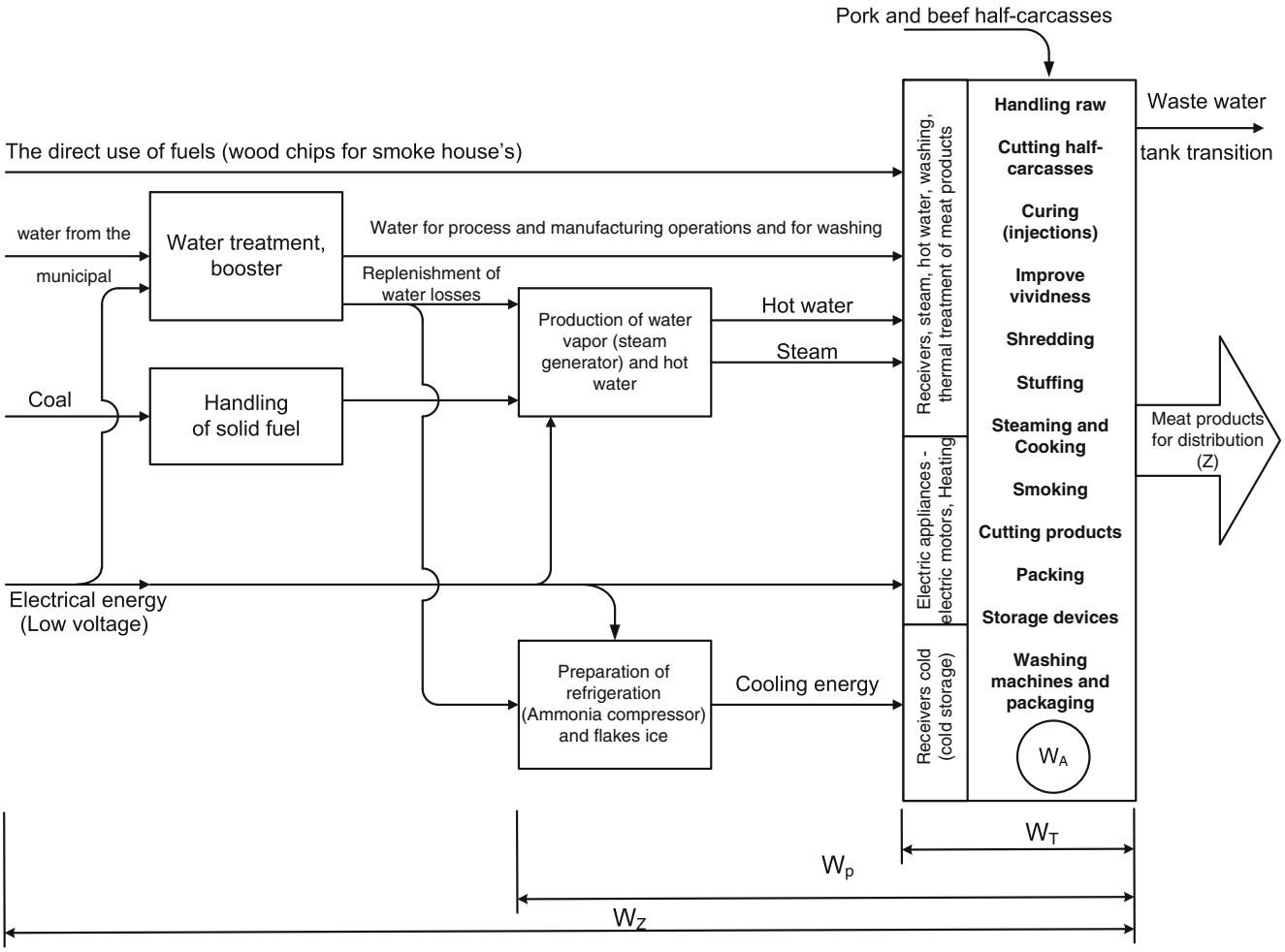


Fig. 1 Material and energy flows in the analysed plant

this aim, the unit energy consumption indicators were adopted (items 6 and 7 in Table 3).

These indicators (W_{t1} and W_{t2}) were converted into unit energy consumption of coal equivalent based on Eqs. 8 and 9 (Table 3).

Indicators of total energy consumption (W_{t1} and W_{t2}) were also converted into energy obtainable from various alternative sources using the equations given (see items 10 and 11 in Table 3). Thus, obtained results are mainly of theoretical importance, since the change of energy carrier is associated with the need to modify the energy conversion system in the boiler house. To express the energy efficiency of meat processing, three sample indicators were applied (see items 12–14, Table 3).

Table 2 Breakdown of power by type of electrical equipment

Item	Equipment	Power installed [kW]
1	Cutters and grinders	40.0
2	Smoke houses, cookers	28.0
3	Cooling machine and ice generator	10.0
4	Electrical boilers	7.0
5	Tumbler	5.5
6	Plant lighting	9.0
7	Other devices up to 5 kW	80.5
8	Total	180.0

Similar research methodology for analysis of the energy consumption in agro-food plants is presented in the cited literature (Wojdalski and Dróżdż 2006; Kaleta and Wojdalski 2008).

It is assumed that the monthly production volume (Z) has an impact on demand for energy carriers in the processing plant. Previous studies have shown that this factor has the greatest relevance because of environmental impact assessment of meat processing plants and determination of the best production techniques available (WS Atkins Int 1998; European Commission 2006; IFC—World Bank Group 2007; Wojdalski et al. 2010; Simpson and Kubicki 1998).

Table 3 Indicators used to characterise meat processing plant

Item	Description	Formula	Units
1	Volume of plant production space per 1 Mg of meat products processed in a month	$K_1 = \frac{V_1}{Z}$	m^3/Mg
2	Total plant volume per 1 Mg of meat products processed in a month	$K_2 = \frac{V_2}{Z}$	m^3/Mg
3	Power of installed electrical equipment per 1 Mg of processed products for a month period	$K_m = \frac{P}{Z}$	kW/Mg
4	Monthly production per plant employee	$K_p = \frac{Z}{n}$	Mg/person
5	Electric energy consumption indicator per 1 Mg of meat products	$W_e = \frac{A_e}{Z}$	kW h/Mg
6	Total unit energy consumption indicator (conversion 1 kWh=12 MJ)	$W_{t1} = \frac{A_{t1}}{Z} = \frac{12 \cdot A_e + B_{rz} \cdot Q_w^r}{Z}$	MJ/Mg of product
7	Total unit energy consumption indicator (conversion 1 kWh=3.6 MJ)	$W_{t2} = \frac{A_{t2}}{Z} = \frac{3.6 \cdot A_e + B_{rz} \cdot Q_w^r}{Z}$	MJ/Mg of product
8	Unit energy consumption - coal equivalent (conversion 1 kWh=12 MJ)	$W_{ce1} = \frac{W_{t1}}{Q_{ce}}$	kg CE/Mg of product
9	Unit energy consumption - coal equivalent (conversion 1 kW·h=3.6 MJ)	$W_{ce2} = \frac{W_{t2}}{Q_{ce}}$	kg CE/Mg of product
10	Total unit energy consumption - real fuel (conversion 1 kWh=12 MJ)	$W_{rz1} = \frac{W_{t1}}{Q_w^r}$	kg of fuel/Mg of product
11	Total unit energy consumption - real fuel (conversion 1 kW·h=3.6 MJ)	$W_{rz2} = \frac{W_{t2}}{Q_w^r}$	kg of fuel/Mg of product
12	Energy efficiency indicator - electricity	$EE_e = \frac{Z}{A_e}$	Mg of product/kWh
13	Energy efficiency indicator - total energy	$EE_{t1} = \frac{Z}{A_{t1}}$	Mg of product/MJ
14	Energy efficiency indicator - coal equivalent	$EE_{ce1} = \frac{Z}{B_{ce1}}$	Mg of product/kg CE

In order to determine the impact of meat production (Z) on the actual energy carriers' consumption (Y), Eq. (1) was applied:

$$Y = b + \alpha Z \tag{1}$$

where Y is the energy carriers' consumption (the dependent variable, e.g., W_e, W_{t1}) and Z is the mass of meat processing (the explanatory variable).

Assumption of border conditions:

$$\alpha Z \geq b \tag{2}$$

and

$$Z > 0 \tag{3}$$

using the obtained regression equations with the border conditions met can partly explain this issue in the analysed meat-processing plant.

The use of regression equations to characterise energy consumption in food plants in different seasons is shown in detail in Muller et al. (2007) and Wojdalski and Dróżdż (2006). The methodology for determining energy consumption of production processes presented by Jekayinfa and Bambgboye (2006) may also be found of use in meat-processing plants.

Results and Discussion

Table 4 shows the ranges of operational and organizational indicators of the analysed plant.

Table 5 shows variability ranges of the plant energy consumption by type of energy carrier.

Using statistical analysis of the collected data we obtained linear regression equations (Table 6) expressing variability in energy consumption. Calculations were performed using the STATISTICA Data Miner + QC Polish v. 9.1.

The resulting regression equations (Table 6) describe the fluctuation of the analysed indicators very well, as evidenced

Table 4 Operational and organizational indicators of the plant

Item	Indicator	Units	Range	
			Minimum	Maximum
1	U_{en}	kW/person	2.46 (average value)	
2	K_m	kW/Mg	0.258	0.629
3	K_p	Mg/person	4.15	9.58
4	K1	m^3 of volume of the plant's production space/Mg	7.08	16.33
5	K2	m^3 of total volume of the plant/Mg	9.06	20.91

Table 5 Energy carriers' consumption in the plant (including meat production)

Dependent variable, energy carriers' consumption indicators	Symbols and units	Range	Average
Electric energy	A_e [MWh/month]	27.71–36.63	33.19
	W_e [kWh/Mg of products]	43.80–101.50	82.10
	EE_e [kg/kWh]	9.85 - 22.83	12.18
Total energy	A_{t1} [GJ/month]	628.46–735.47	694.18
	W_{t1} [MJ/Mg of products]	949.0–2252.0	1723.0
	EE_{t1} [kg of product/MJ]	0.444–1.054	0.580
	A_{t2} [GJ/month]	395.66–427.76	415.616
	W_{t2} [MJ/Mg of product]	581.0–1399.0	1034.0
Coal equivalent	B_{ce1} [Mg CE/month]	21.44–25.09	23.69
	W_{ce1} [kg CE/Mg of product]	32.38–76.84	58.81
	EE_{c1} [kg of product/kg CE]	13.01–30.88	17.00
	B_{ce2} [Mg CE/month]	13.50–14.59	14.18
	W_{ce2} [kg CE/Mg of product]	19.82–47.73	35.27

by the high values of the coefficients of determination (R^2). Large changes in the value (R^2=90 %) of the index of unit energy consumption (W_e) was due to the impact of other, non-production factors not analysed in this work. However, changes of W_e index are similar to the changes in other indicators used to evaluate the energy consumption in the plant.

The increased non-productive energy consumption could have resulted from a relatively large number of small devices, whose share in the total installed power was 44.7 % (Table 2).

The indicators K1 and K2 (rows 4 and 5 in Table 4) provide information concerning the use of production space of the plant. Simpson and Kubicki (1998) showed that in a typical meat plant, cooling systems consume the most power (up to 40 % of the total). Refrigeration compressors and flake ice generators represent only 5 to 25 % of the installed power in meat plants. It should also be noted that many plants do not conduct ongoing analysis of their energy consumption.

Table 6 Impact of monthly production Z (Mg/month) on energy carriers' consumption

Item	Regression equation	R^2 [%]	Units
1	$W_e = 137.226 - 0.131 \cdot Z$	90.0	kWh/Mg
2	$W_{t1} = 2989.926 - 3.015 \cdot Z$	95.5	MJ/Mg
3	$W_{t2} = 1837.234 - 1.913 \cdot Z$	92.9	MJ/Mg
4	$K_m = 0.817 - 0.001 \cdot Z$	87.5	kW/Mg

The results obtained were compared with the data in literature (IFC—World Bank Group 2007). The publication (IFC—World Bank Group 2007) shows that the total unit energy consumption for meat processing plants ranged between 110 and 760 kWh/Mg of pig carcasses (raw material), which, converted into units used in this work, is 0.396–2.763 GJ/Mg. The comparison made shows that energy consumption in the studied plant, as expressed by W_{t2} indicator, was within 0.581–1.399 GJ/Mg of products. Therefore, the energy consumption was below the average for the group of similar plants mentioned in the publication (IFC—World Bank Group 2007).

According to the study (WS Atkins Int. 1998) the electric energy consumption per unit of output (expressed by the indicator W_e) for the Polish meat-processing plants with full production profile was, on average, approximately 1,260 kWh/Mg of products (ranging from 70 to 8,140 kWh/Mg of products). The corresponding values for heat were approximately 24.5 GJ/Mg of products. Skonecki et al. (2001) present data for consumption of selected energy carriers: fuel, 4.311 GJ/Mg of products and water steam, 3.613 GJ/Mg. The paper (European Commission 2006) presents only the energy consumption for selected meat products, including 2,500–4,000 kWh/Mg of ham and 150–180 m^3 of methane/Mg of ham and it does not provide any methodological background for determining the above-mentioned indicators.

The indices presented in Tables 5 and 6 may also be useful in analysing the impact of the manufacturing plant on the environment. This analysis could include a possible reduction in greenhouse gas emissions if the fuel oil used for heat production was replaced by biomass (see alternative energy sources listed in Table 7). In order to obtain environmental benefits, the methodology set out in a publication of the IPCC (1996) would be of use. Results of such analysis could refer to other papers covering the environmental aspects of production and meat

Table 7 Unit energy consumption indicators for different fuels

Energy carriers	Calorific value Q^r_w, [MJ/kg]	Average unit fuel consumption indicators	
		W_{rz1} [kg of fuel/Mg of product]	W_{rz2} [kg of fuel/Mg of product]
Tree briskets and pellets	18.0	95.7	57.4
Straw briskets	17.1	100.8	60.5
Wheat straw	17.3	99.6	59.8
Rape straw	15.0	114.9	68.9
Sawdust	19.3	89.3	53.6
Willow chips	16.5	104.4	62.7

processing (Alcázar-Ortega et al. 2012; Burke et al. 2009; James and James 2010; Nielsen et al. 2008; Roy et al. 2012).

In Poland, electricity production is mainly based on energy from coal. Therefore, any change in unit energy consumption affects the amount of CO_2 emissions. Thermal energy in the analysed plant was obtained from combustion of liquid fuels.

Using the indicators W_{t1} and W_{t2} and taking into account the calorific values of different fuels referred to in literature (Niedziółka and Zuchniarz 2006). Table 7 shows the consumption of different energy carriers (real fuel). For the purpose of calculations, the calorific values, Q_w^r are expressed in megagrams per kilogram of real fuel.

Selected figures contained in Table 7 may be used in the above-mentioned research opportunity for analysing possibilities of replacing fossil fuels by alternative sources.

The W_{rz1} index expresses the average unit demand for a given energy carrier if it fully replaced the primary energy used to produce electric power supplied to the meat plant and heat energy consumed in this plant. The W_{rz2} index has a narrower range and indicates the average unit demand for an energy carrier equivalent only to the energy consumed in the meat plant A_{t2}.

The results presented here may be useful in improving energy management by determining the volume of production at which the indicators W_e, W_{t1} and W_{t2} achieve their lowest values. Moreover, they can be used for comparison with those for other meat processing plants in terms of the specific features arising from production technology, applied research methods and applied technologies such as heat recovery (Fritzson and Berntsson 2006; Kowalczyk and Netter 2008; Simpson and Kubicki 1998). Benchmarking can be used to draw conclusions about possible changes in unit energy consumption in the plant.

Conclusions

The studied meat processing plant was characterised by low values of unit energy carriers' consumption indicators when compared with the literature on plants with a similar profile and processing technology and may be considered as a good model for other similar plants (no slaughter operations, annual production of app. 5,000 Mg of meat products).

The energy consumption of the plants included in Table 1 is 1.5–5 times higher than in the analysed plant. It has been shown in the tested plant that production volume is a fundamental factor conditioning the thermal and electric energy consumption. This study shows the usefulness of introducing active monitoring of energy carriers' consumption as one of the best energy management techniques when used in line with existing volumes of production. The regression equations obtained allow determining the monthly production volume at which the specific energy consumption indicators are the lowest.

The results contained in the present study may also be useful for verifying environmental standards. Since the obtained results relate to the specific conditions of production and the specific level of utilization of installed electrical equipment, they may be of use in estimating production costs and air emissions in comparable processing plants. It was the aim of this paper to present a methodology which, due to its comprehensiveness, has a potential to be used in benchmarking procedures allowing for a more detailed comparison of plants as energo-technological systems.

Further research should lead to answering the question what is the optimal energy consumption for a specific type of meat plant when taking into account its utilization of the installed power, production area and the number of employees. In addition, studies should include the issues of efficiency of energy transformation and energy consumption in the largest groups of production equipment using the W_A, W_T and W_P indices defined in this paper. Research may also include the importance of water as a heat carrier.

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References

Alcázar-Ortega, M., Álvarez-Bel, C., Escrivá-Escrivá, G., & Domijan, A. (2012). Evaluation and assessment of demand response potential applied to the meat industry. *Applied Energy, 92*, 84–91.

Banach, J., & Żywica, R. (2010). The effect of electrical stimulation and freezing on electrical conductivity of beef trimmed at various times after slaughter. *Journal of Food Engineering, 100*, 119–124.

Bianchi, M., Cherubini, F., De Pascale, A., Peretto, A., & Elmegaard, B. (2006). Cogeneration from poultry industry wastes: indirectly fired gas turbine application. *Energy, 31*, 1417–1436.

Bieranowski, J., & Klonowski, A. (2005). The model of an associated energy source in food processing industry (in Polish: Model systemu skojarzonego źródła energii cieplnej w zakładzie przemysłu rolno-spożywczego). *Inżynieria Rolnicza, 6*, 33–40.

Budny, J., Piechocki, J., & Piskunowicz, W. (1987). Power consumption during process of cooling pork and beef meat (in Polish: Energochłonność wychładzania półtusz wołowych i wieprzowych). *Chłodnictwo, 10*, 20–22.

Bunse, K., Vodicka, M., Schönsleben, P., Brülhart, M., & Ernst, F. O. (2011). Integrating energy efficiency performance in production management—gap analysis between industrial needs and scientific literature. *Journal of Cleaner Production, 19*, 667–679.

Burke, M., Oleson, K., McCullough, E., & Gaskell, J. (2009). A global model tracking water, nitrogen, and land inputs and virtual transfers from industrialized meat production and trade. *Environmental Modeling and Assessment, 2*(14), 179–193.

Campañone, A., Analía, L., Zaritzky, A., & Noemi, E. (2010). Mathematical modeling and simulation of microwave thawing of large

solid foods under different operating conditions. *Food and Bioprocess Technology, 6*, 813–825.

Cierach, M., Białobrzewski, I., & Markowski, M. (2000). Investigation of meat products heating and cooling processes (in Polish: Badania procesów ogrzewania i chłodzenia przetworów mięsnych). *Inżynieria Rolnicza., 5*(16), 39–46.

Dolata, W. (1992). The influence of some technical parameters of cutter on meat batters and sausages quality and energy consumption during comminution (in Polish: Wpływ niektórych parametrów technicznych kutra na kształtowanie jakości farszów i wędlin oraz energochłonności procesu kutrowania). *Roczniki Akademii Rolniczej w Poznaniu, Zeszyt, 225*, 65–85.

Dróżdż, B. (2010). *Energy and water consumption in poultry plants (in Polish: Zużycie energii i wody w zakładach przetwórstwa drobiarskiego). Technological Progress in Food Processing (in Polish: Postępy Techniki Przetwórstwa Spożywczego)* (1st ed., pp. 41–46). Warszawa: Wyższa Szkoła Menedżerska.

European Commission. (2006). Integrated pollution prevention and control. *Reference Document on Best Available Techniques in the Food, Drink and Milk Industries, 149–151*, 155.

Fritzson, A., & Berntsson, Th. (2006). Energy efficiency in the slaughter and meat processing industry—opportunities for improvements in future energy markets. *Journal of Food Engineering, 77*, 792–802.

Gogate, P. R. (2011). Hydrodynamic cavitation for food and water processing. *Food and Bioprocess Technology, 6*, 996–1011.

Grochowicz J & Walczyński S (2004): The Effect of Some Technological Factors on the Energy Consumption of Energy at Mixing Granular Materials. TEKA Commission of Motorization and Power Industry in Agriculture. Polish Academy of Sciences Branch in Lublin. Lublin, 2004, vol. IV, 70–75.

Hansen, P. I., Christiansen, K., & Hummelmose, B. (2000). *Cleaner production assessment in meat processing*. Denmark: COWI.

Houska, M., Sun, D.-W., Landfeld, A., & Zhang, Z. (2003). Experimental study of vacuum cooling of cooked beef in soup. *Journal of Food Engineering, 59*, 105–110.

IFC—World Bank Group (2007) Environmental, Health, and Safety Guidelines for Meat Processing, April 30, 7–24

IPCC (1996) Revised 1996 IPCC Guidelines for National Greenhouse Gas Inventories.

James, S. J., & James, C. (2010). The food cold-chain and climate change. *Food Research International, 7*, 43,1944–43,1956.

Jekayinfa, S. O. (2007). Energetic analysis of poultry processing operations. *Leonardo Journal of Sciences, 10*, 77–92.

Jekayinfa, S. O., & Bamgboye, A. I. (2006). Estimating energy requirement in cashew (*Anacardium occidentale* L.) nut processing operations. *Energy, 31*, 1305–1320.

Kaleta A., Wojdalski J. (ed.). 2008. Food processing. Selected issues—engineering and energy production (in Polish: Przetwórstwo rolno-spożywcze. Wybrane zagadnienia inżynieryjno – produkcyjne i energetyczne). Wydawnictwo SGGW. Warszawa, 182-192.

Klemeš J., Smith R., Kim J-K., 2008. Handbook of water and energy management in food processing. CRC–Cambridge Woodhead.

Kowalczyk, R., & Netter, J. (2008). A new look at the energetic factors consumption in food industry (in Polish: Nowe spojrzenie na zużycie czynników energetycznych w zakładzie przemysłu spożywczego. *Postępy Techniki Przetwórstwa Spożywczego, 1*, 45–47.

Li, C. B., Zhou, G. H., & Xu, X. L. (2010). Dynamical changes of beef intramuscular connective tissue and muscle fiber during heating and their effects on beef shear force. *Food and Bioprocess Technology, 3*, 521–527.

Marcotte, M., Taherian, A. R., & Karimi, Y. (2008). Thermophysical properties of processed meat and poultry products. *Journal of Food Engineering, 88*, 315–322.

Markowski, M., Białobrzewski, I., Cierach, M., & Paulo, M. (2004). Determination of thermal diffusivity of Lyoner type sausages during water bath cooking and cooling. *Journal of Food Engineering, 65*, 591–598.

Muller, D. C. A., Marechal, F. M. A., Wolewinski, T., & Roux, P. J. (2007). An energy management method for the food industry. *Applied Thermal Engineering, 27*, 2677–2686.

Neryng A, Wojdalski J, Budny J, Krasowski E (1990) Energy and water in agro-food industry. WNT, Warszawa, 99-103, 184-189. (in Polish: Energia i woda w przemyśle rolno-spożywczym);

Niedziółka, I., & Zuchniarz, A. (2006). *An energetic analysis of selected plant biomass samples (in Polish: Analiza energetyczna wybranych rodzajów biomasy pochodzenia roślinnego)* (pp. 232–237). Lublin: MOTROL, Motoryzacja i Energetyka Rolnictwa. Tom 8A.

Nielsen, P. H., Dalgaard, R., Korsbak, A., & Pettersson, D. (2008). Environmental assessment of digestibility improvement factors applied in animal production. *The International Journal of Life Cycle Assessment, 13*, 49–56.

Norton, T., & Sun, D.-W. (2008). Recent advances in the use of high pressure as an effective processing technique in the food industry. *Food and Bioprocess Technology, 1*(1), 2–34.

Pagan R, Renouf M, Prasad P (2002) Eco-efficiency manual for meat processing. Meat and Livestock Australia Ltd.

Pimentel, D., Williamson, S., Alexander, C. E., Gonzalez-Pagan, O., Kontak, C., & Mulkey, S. E. (2008). Reducing energy inputs in the US food system. *Human Ecology, 36*, 459–471.

Podfilipski, J., & Żelazny, J. (1985). Power consumption economy applied to technology in meat processing factories—selected problems (in Polish: Gospodarka energetyczna w procesach technologicznych zakładów mięsnych—wybrane zagadnienia). *Chłodnictwo, 5*, 14–18.

Ramirez, C. A., Patel, M., & Blok, K. (2006). How much energy to process one pound of meat? A comparison of energy use and specific energy consumption in the meat industry of four European countries. *Energy, 31*, 2047–2063.

Roy, P., Orikasa, T., Thammawong, M., Nakamura, N., Xu, Q., & Shiina, T. (2012). Life cycle of meats: an opportunity to abate the greenhouse gas emission from meat industry in Japan. *Journal of Environmental Management, 1*(93), 218–224.

Simpson, A., & Kubicki, M. (1998). *Environmental protection in the meat industry (in Polish: Ochrona środowiska w przemyśle mięsnym)* (pp. 42–45). Warszawa: FAPA.

Singh R P (ed.) (1986) Energy accounting of food processing operations (in Energy in Food Processing. Elsevier. Amsterdam 26).

Skonecki, S., Laskowski, J., & Szymczak, W. (2001). *Analysis of the energy consumption in meat plant* (pp. 326–332). Lublin: MOTROL, Motoryzacja i Energetyka Rolnictwa. Tom IV. in Polish: Analiza zużycia energii w zakładzie mięsnym.

Swords, B., Coyle, E., & Norton, B. (2008). An enterprise energy-information system. *Applied Energy, 85*, 61–69.

Tkacz, K. (2002). Energy consumption in cuttering of meat raw products. *Technica Agraria, 1*(2), 73–79 (in Polish Energochłonność kutrowania surowców mięsnych).

Tkacz, K., & Iwaniak, A. (2001). Electric energy consumption by manufacturing of selected ready to cook food products. *Inżynieria Rolnicza, 10*, 371–378. in Polish: Energochłonność elektryczna produkcji wybranych wyrobów garmażeryjnych.

Tkacz, K., Budny, J., & Borowski, J. (2000). Energetic characteristics of beef steak thermal processing. *Inżynieria Rolnicza, 5*(16), 241–248. in Polish: Charakterystyka energetyczna obróbki cieplnej mięsa wołowego.

Wang, L. J. (ed) 2008. Energy efficiency and management in food processing facilities. CRC Press, Boca Raton, FL, USA. ISBN: 1420063383.

Wilhite, H. (2008). New thinking on the agentive relationship between end-use technologies and energy-using practices. *Energy Efficiency, 2*, 121–130.

Wojdalski, J., & Dróżdż, B. (2006). *A rudimental analysis of energy consumption during production at agri-food industry work. (in Polish: Podstawy analizy energochłonności produkcji zakładów przemysłu rolno-spożywczego)* (pp. 294–304). Lublin: MOTROL, Motoryzacja i Energetyka Rolnictwa. Tom 8A.

Wojdalski J, Dróżdż B, Powęzka A (2009) Effectiveness of energy and water consumption in a poultry processing plant. TEKA Commission of Motorization and Power Industry in Agriculture. Polish Academy of Sciences Branch in Lublin. Lublin, 2009, vol. IX, 395-402.

Wojdalski J, Dróżdż B, Lipiński P (2010) Energy consumption efficiency in a small meat processing plant. Współczesne zagadnienia rozwoju sektora energetycznego i rolniczego. SGGW, 110-121 (in Polish: Efektywność zużycia energii w małym zakładzie przetwórstwa mięsnego).

WS Atkins Int. (1998). *Environmental protection in the agro-food industry. Environmental Standards (in Polish: Ochrona środowiska w przemyśle rolno-spożywczym. Standardy środowiskowe)* (31st ed.). Warszawa: FAPA. 78, 87, 106-107.